

HARI GOWTHAM ARETI

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PROFESSIONAL SUMMARY

Computer Science undergraduate (graduating 2026, CGPA 8.82) with hands-on experience in Python ETL pipelines, ML model development, and REST API deployment. Developing working knowledge of LLM-powered workflows, agentic patterns, RAG pipelines, and tool-integrated systems. Seeking to apply software engineering fundamentals and AI skills to production-grade agentic AI and MCP system development.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, SQL, Flask, REST APIs, SQLAlchemy, HTML/CSS
- **AI / ML:** TensorFlow, Scikit-learn, OpenCV, EfficientNet, CNN, Transfer Learning, LLM Prompt Design, RAG Concepts, Tool/Function Calling, Embedding & Vector Search (conceptual), Agent Orchestration Patterns
- **Data Engineering:** Python ETL Pipelines, Pandas, NumPy, Feature Engineering, Data Preprocessing, Data Validation
- **BI & Visualization:** Power BI (DAX, Power Query), Excel (Pivot Tables, Charts), Matplotlib, Tableau
- **Tools & Platforms:** Jupyter Notebook, Git, GitHub, Postman, VS Code, Cursor (AI-powered IDE)
- **Software Engineering:** SDLC, Modular Code Architecture, Code Documentation, API Integration, Microservices, CI/CD Basics

WORK EXPERIENCE

AI/ML Engineering Intern

Jan 2025 – Apr 2025

Blackbuck Engineers

- Built Python ETL pipelines for data collection, transformation, and loading, improving processing efficiency by 25%.
- Designed and deployed ML inference REST APIs using Flask, enabling client-side integration and microservice-based architecture.
- Performed data preprocessing, feature engineering, and model validation to support end-to-end ML model development.
- Analyzed model performance metrics (Accuracy, F1-Score, AUC-ROC) and maintained structured documentation of experimental results.

PROJECTS

Sign Gesture Detection | Python · TensorFlow · OpenCV · Scikit-learn · CNN | GitHub

- Built a CNN-based real-time sign gesture recognition system using OpenCV for video capture and frame-level inference.
- Applied data augmentation and transfer learning to improve generalization; evaluated using F1-Score, Precision, and Recall.

Deepfake Image Detection | Python · TensorFlow · EfficientNet · OpenCV · Scikit-learn | GitHub

- Built a CNN-based deepfake classifier using EfficientNet transfer learning with OpenCV preprocessing, achieving 91%+ accuracy.
- Evaluated model robustness using F1-Score, Precision, Recall, and AUC-ROC with iterative error analysis.

Chest X-Ray Diagnosis | Python · TensorFlow · OpenCV · Deep Learning · Medical Imaging | GitHub

- Developed a deep learning model to classify chest X-ray images for pneumonia detection using CNN and transfer learning.
- Evaluated performance using AUC-ROC, Precision, and Recall on imbalanced medical imaging datasets.

EDUCATION

B.Tech in Computer Science Engineering

2022 – 2026

Swarnandhra College of Engineering and Technology

CGPA: 8.82 / 10

CERTIFICATIONS

- Google Data Analytics Professional Certificate – Coursera
- AI/ML Engineering Intern Certificate – Blackbuck Engineers